An Empirical Comparison of Five Classifiers on Four UCI Datasets

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Abstract

I empirically compare five supervised learning algorithms (Random Forest, Gradient Boosting, Support Vector Machine with RBF kernel, Logistic Regression, and k-Nearest Neighbors) on four binary classification datasets from the UCI repository: Breast Cancer Wisconsin (Diagnostic), Spambase, Adult Income, and Bank Marketing. Following the evaluation methodology of Caruana and Niculescu-Mizil (2006), I vary the training fraction (20/80, 50/50, 80/20), perform cross-validated hyperparameter tuning on each (dataset, classifier, partition) combination, and evaluate performance on held-out test sets. I report both raw accuracies and CNM-style normalized scores relative to a majority-class baseline and the best observed classifier on each dataset. The results confirm several trends from prior work: tree-based ensembles consistently perform best, k-NN is competitive but rarely state-of-the-art, and increasing training data almost always improves test accuracy, especially on more difficult and imbalanced datasets.

1. Introduction

Supervised learning algorithms differ widely in their inductive biases, representational power, and robustness to noise and class imbalance. Consequently, their empirical performance can vary substantially across tasks, even when trained and tuned carefully. This point was made forcefully by Caruana & Niculescu-Mizil (2006), who performed a large-scale comparison of ten supervised learning algorithms on a collection of real-world benchmark problems using multiple evaluation metrics. Their study showed that there is no single universally best classifier, but also that some families of models (especially boosted and bagged decision trees) tend to perform well across many datasets.

Motivated by this line of work, I carry out a smaller-scale but systematic empirical comparison of five widely used supervised learning algorithms on four benchmark datasets from the UCI Machine Learning Repository. The classifiers I study are:

- Random Forests,
- Gradient Boosting (decision trees),
- Support Vector Machines with RBF kernel (SVM-RBF),
- Logistic Regression, and
- k-Nearest Neighbors (k-NN).

These models span several major paradigms in classical machine learning: ensemble tree methods, kernel-based margin maximization, linear discriminative models, and instance-based nonparametric classification. All are implemented using the same software stack and trained within a common preprocessing and cross-validation framework, which enables fairer comparison of their strengths and weaknesses.

To probe how algorithm rankings depend on data characteristics, I evaluate each classifier on four binary classification datasets that differ substantially in size, feature types, and class balance: Breast Cancer Wisconsin (Diagnostic), Spambase, Adult Income, and Bank Marketing. For each (dataset, classifier) pair I vary the training fraction (20/80, 50/50, 80/20 train/test splits), perform stratified cross-validation to select hyperparameters, and then evaluate the chosen model on held-out test data. This yields

$$\begin{aligned}&5 \text{ classifiers} \times 4 \text{ datasets} \times \\ &3 \text{ splits} \times 3 \text{ random trials} = 180 \text{ independent runs.}\end{aligned}$$

Following Caruana & Niculescu-Mizil (2006), I focus primarily on classification accuracy but also report F_1-score and ROC AUC to better capture performance on imbalanced datasets. In addition, I compute CNM-style normalized scores that place each classifier between a trivial majority baseline and the best observed classifier on each dataset, allowing us to aggregate results across datasets with different difficulty. The goal is not to reproduce the exact numbers from the original study, but to see whether similar qualitative trends emerge when using modern implementations and a smaller set of representative algorithms and datasets.

2. Datasets and Problem Setup

I use four binary classification datasets from the UCI Machine Learning Repository (Caruana & Niculescu-Mizil, 2006; ?). Together they cover small to large sample sizes, purely numeric as well as mixed numeric–categorical feature spaces, and varying degrees of class imbalance. All tasks are cast as two-class problems: positive vs negative class.

2.1. Breast Cancer Wisconsin (Diagnostic)

The Breast Cancer Wisconsin (Diagnostic) dataset contains measurements derived from digitized images of fine needle aspirates (FNA) of breast masses. Each instance corresponds to a tumor and is labeled as benign or malignant.

- **Domain:** Medical / oncology
- **Instances:** 569
- **Features:** 30 real-valued attributes
- **Classes:** Benign (62%), Malignant (38%)
- **Properties:** No missing values, low noise, purely numeric features.

This dataset is relatively small and clean, making it useful for studying overfitting, model flexibility, and performance when only limited training data are available.

2.2. Spambase

The Spambase dataset represents email messages using hand-crafted statistics over words, characters, and capitalization patterns. The task is to distinguish spam from non-spam ("ham") messages.

- **Domain:** Email / text-derived numeric features
- **Instances:** 4,601
- **Features:** 57 continuous attributes
- **Classes:** Non-spam ($\approx 61\%$), Spam ($\approx 39\%$)
- **Properties:** Medium-sized, all numeric, derived from text statistics; requires feature scaling for distance-based and kernel methods.

Compared to Breast Cancer, this dataset is higher-dimensional and noisier, and is a good testbed for algorithms that can exploit many weakly informative features.

2.3. Adult Income

The Adult Income ("Census Income") dataset is built from US census data. The goal is to predict whether an individual's annual income exceeds \$50K.

- **Domain:** Socio-economic / census
- **Instances:** 48,842
- **Features:** 14 attributes (a mix of numerical and categorical; one-hot encoding yields $\sim$100 derived features)
- **Classes:** $\leq$ \$50K ($\approx 76\%$), $>$ \$50K ($\approx 24\%$)
- **Properties:** Large-scale, mixed-type features, moderate class imbalance, some missing values encoded as "?".

This dataset allows us to study how the classifiers scale to larger sample sizes and how sensitive they are to class imbalance and categorical feature handling.

2.4. Bank Marketing

The Bank Marketing dataset contains information from direct marketing campaigns (telephone calls) run by a Portuguese banking institution. The task is to predict whether a client will subscribe to a term deposit.

- **Domain:** Finance / marketing
- **Instances:** 45,211
- **Features:** 16 attributes (mostly categorical; a few numerical)
- **Classes:** No subscription ($\approx 88\%$), Subscription ($\approx 12\%$)
- **Properties:** Strong class imbalance, extensive categorical encoding, and potentially complex interactions between demographic and campaign features.

The heavy skew toward the negative class makes this dataset challenging for accuracy alone and highlights the value of metrics such as F_1 and ROC AUC.

2.5. Dataset Summary

Table 1 summarizes the key characteristics of the four datasets. Together they span small to large sample sizes, low- to moderate-dimensional feature spaces, pure numeric vs mixed feature types, and near-balanced vs severely imbalanced class distributions. This diversity is important for evaluating whether algorithm rankings are stable across different data regimes.

Table 1. Summary of the four UCI datasets used in this study. The positive class corresponds to malignant tumors (Breast Cancer), spam emails (Spambase), high income (Adult), and term-deposit subscription (Bank).

Dataset	Instances	Features	Types	Pos. rate
Breast Cancer	569	30	Num.	38%
Spambase	4,601	57	Num.	39%
Adult Income	48,842	$\sim$100	Mixed (enc.)	24%
Bank Marketing	45,211	$\sim$100	Mostly cat.	12%

In all experiments I treat each dataset as a binary classification problem and use the original train–test splits only to obtain the raw data; all actual training and evaluation splits are created programmatically as described in Section 3.

3. Methods

This section describes the classifiers, preprocessing, and evaluation protocol used in the experiments. To facilitate fair comparison, all methods are implemented in Python using the same preprocessing pipelines and cross-validation strategy.

3.1. Classifiers

I evaluate five supervised learning algorithms that are commonly used for classification on tabular data.

Random Forest (RF). Random Forests are ensembles of decision trees trained on bootstrap samples of the data, with random feature subsampling at each split (**?**). They can model nonlinear decision boundaries and feature interactions, handle heterogeneous feature types, and are relatively robust to noise.

Gradient Boosting (GB). Gradient Boosting builds an additive ensemble of shallow decision trees, where each new tree is trained to correct the residual errors of the current ensemble. With appropriate regularization (learning rate, tree depth, and number of trees), gradient boosting can capture complex patterns while controlling overfitting.

Support Vector Machine with RBF kernel (SVM-RBF). Support Vector Machines learn a maximum-margin separating hyperplane in a feature space induced by a kernel function. I use the radial basis function (RBF) kernel, which can approximate complex nonlinear boundaries. SVMs are sensitive to the choice of the regularization parameter C and kernel width γ, and require standardized features.

Logistic Regression (LR). Logistic Regression is a linear discriminative model that estimates class probabilities via the logistic function. With ℓ_2 regularization it can handle moderate-dimensional feature spaces and serves as a strong baseline, especially when the true decision boundary is approximately linear in the transformed feature space.

k-Nearest Neighbors (k-NN). k-NN is a nonparametric, instance-based method that predicts the class of a query point from the labels of its k closest training examples. It makes few assumptions about the data distribution but is sensitive to feature scaling and can perform poorly in high dimensions.

3.2. Experimental Protocol

All experiments are implemented in Python using `scikit-learn` for the classifiers and model-selection tools, `pandas` and `numpy` for data handling, and `matplotlib` for visualization. The UCI datasets are loaded via the `ucimlrepo` package, which provides convenient access to the original feature and target tables.

Data loading and cleaning. For each dataset I load the feature matrix and target vector from `ucimlrepo` and apply minimal cleaning. For Adult, I replace "?" entries with missing values and drop rows with missing attributes before splitting into features and labels. For Bank Marketing, I map the "yes"/"no" term-deposit outcome to $\{1, 0\}$; for Breast Cancer and Spambase I similarly convert the original labels to binary $\{0, 1\}$. No additional feature engineering is performed beyond the generic preprocessing described below.

Preprocessing. Each dataset is preprocessed using a `ColumnTransformer` with separate pipelines for numeric and categorical features:

- Numeric attributes are imputed with the median and then standardized using `StandardScaler`.
- Categorical attributes are imputed with the most frequent value and one-hot encoded (`OneHotEncoder` with `handle_unknown="ignore"`), ignoring unseen categories at test time.

The same preprocessing transformer is used for all five classifiers on a given dataset and training partition, so performance differences arise from the learning algorithms rather than from feature engineering.

Train/test partitions and trials. For each dataset I consider three training fractions: 20%, 50%, and 80%, corresponding to 20/80, 50/50, and 80/20 train/test splits. For each fraction I generate three independent stratified splits using different random seeds. This yields three test sets per (dataset, training fraction), and I report mean performance across these trials to reduce variance due to a single lucky or unlucky split.

Hyperparameter tuning. For every combination of dataset, classifier, and training fraction I perform hyperparameter selection using stratified k-fold cross-validation on the training portion only. I wrap the preprocessor and classifier in a single `Pipeline` of the form `[("preprocess", ColumnTransformer), ("model", Estimator)]` and run `GridSearchCV` with a modest grid of hyperparameters specific to each classifier:

- **Random Forest (RF):**
  `n_estimators` $\in \{100, 300\}$; `max_depth` $\in \{\text{None}, 5, 10\}$; `min_samples_leaf` $\in \{1, 5\}$; `max_features` $\in \{\text{"sqrt"}, 0.5\}$.
- **Gradient Boosting (GB):**
  `n_estimators` $\in \{100, 300\}$; `learning_rate` $\in \{0.05, 0.1, 0.2\}$; `max_depth` $\in \{1, 3\}$; `subsample` $\in \{0.8, 1.0\}$.
- **SVM with RBF kernel (SVM-RBF):**
  regularization parameter `C` $\in \{0.1, 1, 10\}$; kernel width `gamma` $\in \{0.01, 0.1, 1\}$. I use the `SVC` implementation with probability estimates enabled to compute ROC AUC.
- **Logistic Regression (LR):**
  inverse regularization strength `C` $\in \{0.01, 0.1, 1, 10\}$; `penalty` fixed to ℓ_2; `solver = "lbfgs"` with a maximum of 1,000 iterations.
- **k-Nearest Neighbors (k-NN):**
  number of neighbors `n_neighbors` $\in \{3, 5, 11, 21\}$; `weights` $\in \{\text{"uniform"}, \text{"distance"}\}$; `metric` fixed to Euclidean distance.

These ranges are chosen to cover a mix of under- and over-regularized settings without making the search prohibitively expensive. I use 5-fold stratified cross-validation for small and medium-sized datasets and reduce to 3-folds for the largest training partitions to keep runtime manageable. The best setting is selected by mean cross-validated accuracy and recorded along with its validation score.

Evaluation metrics and logging. Once the best hyperparameters have been selected, I refit the pipeline on the full training set and evaluate on the held-out test set. For each run I record:

1. training accuracy (to check for underfitting / overfitting)
2. cross-validated validation accuracy (from the grid search)
3. test accuracy, test F_1-score (positive class)
4. test ROC AUC (when probabilistic scores are available)

Accuracy is the primary metric, matching the focus of Caruana & Niculescu-Mizil (2006), but F_1 and ROC AUC are especially informative for the imbalanced Adult and Bank Marketing datasets.

In addition, I compute CNM-style normalized scores to compare performance across datasets of differing difficulty. For each dataset d I define $\text{baseline}(d)$ as the accuracy of a trivial majority-class predictor and $\text{best}(d)$ as the highest mean test accuracy achieved by any of the five classifiers. The normalized score for classifier c on dataset d is

$$\text{norm\_score}(d, c) = \frac{\text{acc}(d, c) - \text{baseline}(d)}{\text{best}(d) - \text{baseline}(d)}.$$

I then average these normalized scores across the four datasets to obtain a single CNM-style score per classifier, following the normalization used in Caruana & Niculescu-Mizil (2006).

4. Experiments and Results

I now summarize the empirical behavior of the five classifiers across the four datasets, three training fractions, and three random trials per fraction. Unless otherwise stated, all reported numbers are averages over the three trials.

4.1. Per-dataset, Per-Partition Comparisons

For each dataset and each train/test partition, I aggregate the `all_results` table to obtain mean and standard deviation of training accuracy, cross-validated accuracy, and test accuracy for every classifier. These values are reported in Tables below.

For each table, CV Acc = cross-validated validation accuracy. RF = Random Forest, GB = Gradient Boosting, LR = Logistic Regression. Values are mean $\pm$ std over three trials.

Breast Cancer. All methods achieve very high test accuracy (≈ 0.93–0.97) for every partition, so the dataset is close to ceiling performance. Logistic Regression and RBF-SVM are usually the top performers, with test accuracy around 0.96–0.97 across splits, while the two ensemble methods (Gradient Boosting and Random Forests) and KNN trail by only 1–2 percentage points. Training accuracy is near 1.0 for most models, but the small train/test gap indicates limited overfitting. Overall, Breast Cancer behaves like a nearly linearly separable problem on which all reasonable classifiers perform similarly.

Table 2. Per-partition performance on the Breast Cancer dataset.

Split	Classifier	CV Acc	Train Acc	Test Acc
20/80	GB	0.964 ± 0.000	0.994 ± 0.010	0.925 ± 0.011
20/80	KNN	0.964 ± 0.000	0.973 ± 0.015	0.942 ± 0.028
20/80	LR	0.982 ± 0.000	1.000 ± 0.000	0.960 ± 0.019
20/80	RF	0.982 ± 0.000	0.991 ± 0.000	0.930 ± 0.019
20/80	SVM	0.964 ± 0.000	0.991 ± 0.009	0.959 ± 0.011
50/50	GB	0.982 ± 0.000	1.000 ± 0.000	0.956 ± 0.005
50/50	KNN	0.958 ± 0.000	1.000 ± 0.000	0.951 ± 0.007
50/50	LR	0.979 ± 0.000	0.993 ± 0.004	0.971 ± 0.004
50/50	RF	0.972 ± 0.000	0.996 ± 0.000	0.949 ± 0.004
50/50	SVM	0.979 ± 0.000	0.996 ± 0.000	0.961 ± 0.007
80/20	GB	0.974 ± 0.000	1.000 ± 0.000	0.953 ± 0.013
80/20	KNN	0.967 ± 0.000	0.979 ± 0.005	0.953 ± 0.013
80/20	LR	0.978 ± 0.000	0.987 ± 0.000	0.968 ± 0.010
80/20	RF	0.963 ± 0.000	1.000 ± 0.000	0.950 ± 0.013
80/20	SVM	0.980 ± 0.000	0.989 ± 0.002	0.965 ± 0.009

Spambase. On Spambase the ordering is clearer. For every partition, Gradient Boosting and Random Forests form the leading tier, with test accuracies around 0.93 (20/80), 0.95 (50/50), and 0.95 (80/20), and Random Forests very slightly ahead on the 50/50 and 80/20 splits. RBF-SVM and Logistic Regression occupy a middle tier, roughly 1–3 points behind the ensembles, while KNN is consistently the worst performer by 3–5 points. KNN and Random Forests often reach near-perfect training accuracy, indicating more overfitting than the linear and kernel methods, but the ensembles still generalize best on this medium-sized numeric dataset.

Table 3. Per-partition performance on the Spambase dataset. Abbreviations as in Table 2.

Split	Classifier	CV Acc	Train Acc	Test Acc
20/80	GB	0.930 ± 0.000	0.974 ± 0.002	0.930 ± 0.002
20/80	KNN	0.874 ± 0.000	1.000 ± 0.000	0.894 ± 0.008
20/80	LR	0.900 ± 0.000	0.935 ± 0.007	0.914 ± 0.004
20/80	RF	0.933 ± 0.000	0.964 ± 0.003	0.928 ± 0.003
20/80	SVM	0.911 ± 0.000	0.966 ± 0.005	0.919 ± 0.003
50/50	GB	0.948 ± 0.000	0.976 ± 0.001	0.949 ± 0.005
50/50	KNN	0.904 ± 0.000	1.000 ± 0.000	0.908 ± 0.005
50/50	LR	0.927 ± 0.000	0.929 ± 0.002	0.923 ± 0.004
50/50	RF	0.946 ± 0.000	1.000 ± 0.000	0.950 ± 0.004
50/50	SVM	0.927 ± 0.000	0.960 ± 0.003	0.932 ± 0.001
80/20	GB	0.951 ± 0.000	0.995 ± 0.001	0.952 ± 0.003
80/20	KNN	0.915 ± 0.000	0.999 ± 0.000	0.924 ± 0.010
80/20	LR	0.919 ± 0.000	0.933 ± 0.003	0.932 ± 0.003
80/20	RF	0.952 ± 0.000	0.999 ± 0.000	0.954 ± 0.003
80/20	SVM	0.931 ± 0.000	0.960 ± 0.001	0.943 ± 0.004

Adult Income. For Adult, Gradient Boosting is the best method on every train/test partition, with test accuracies of about 0.87 (20/80), 0.87 (50/50), and 0.88 (80/20). Random Forests are consistently second, typically 1–2 points behind GB. RBF-SVM and Logistic Regression perform very similarly to one another and lag the tree ensembles by another 1–2 points. KNN is always the weakest model and shows clear overfitting: training accuracy approaches 1.0 while test accuracy remains in the low 0.83–0.84 range. These results align with the expectation that tree-based ensembles handle mixed numeric/categorical census features and moderate class imbalance better than simpler baselines.

Table 4. Per-partition performance on the Adult Income dataset. Abbreviations as in Table 2.

Split	Classifier	CV Acc	Train Acc	Test Acc
20/80	GB	0.865 ± 0.000	0.891 ± 0.005	0.867 ± 0.000
20/80	KNN	0.833 ± 0.000	0.852 ± 0.003	0.833 ± 0.001
20/80	LR	0.851 ± 0.000	0.847 ± 0.004	0.848 ± 0.001
20/80	RF	0.858 ± 0.000	0.894 ± 0.003	0.857 ± 0.002
20/80	SVM	0.851 ± 0.000	0.865 ± 0.003	0.851 ± 0.001
50/50	GB	0.867 ± 0.000	0.891 ± 0.001	0.869 ± 0.001
50/50	KNN	0.834 ± 0.000	1.000 ± 0.000	0.834 ± 0.002
50/50	LR	0.848 ± 0.000	0.849 ± 0.001	0.848 ± 0.000
50/50	RF	0.858 ± 0.000	0.878 ± 0.001	0.859 ± 0.001
50/50	SVM	0.853 ± 0.000	0.864 ± 0.001	0.852 ± 0.003
80/20	GB	0.870 ± 0.000	0.884 ± 0.001	0.875 ± 0.001
80/20	KNN	0.837 ± 0.000	0.858 ± 0.001	0.841 ± 0.003
80/20	LR	0.848 ± 0.000	0.848 ± 0.001	0.851 ± 0.003
80/20	RF	0.861 ± 0.000	0.911 ± 0.000	0.862 ± 0.001
80/20	SVM	0.853 ± 0.000	0.861 ± 0.000	0.856 ± 0.002

Bank Marketing. The Bank dataset shows the strongest and most stable advantage for the ensemble methods. Gradient Boosting is the top performer on all three splits, with test accuracy roughly 0.90 (20/80) and 0.91 (50/50 and 80/20). Random Forests are a close second, within about 0.3 percentage points of GB on each partition. Logistic Regression and RBF-SVM are slightly worse but still competitive, while KNN is again the weakest classifier with test accuracy around 0.895–0.897. Despite very high training accuracy for the ensembles (especially Random Forests), their generalization gap remains small, suggesting effective regularization via shallow trees and subsampling.

Table 5. Per-partition performance on the Bank Marketing dataset. Abbreviations as in Table 2.

Split	Classifier	CV Acc	Train Acc	Test Acc
20/80	GB	0.901 ± 0.000	0.938 ± 0.002	0.904 ± 0.001
20/80	KNN	0.894 ± 0.000	0.904 ± 0.002	0.895 ± 0.001
20/80	LR	0.899 ± 0.000	0.904 ± 0.003	0.901 ± 0.000
20/80	RF	0.898 ± 0.000	0.991 ± 0.001	0.902 ± 0.001
20/80	SVM	0.897 ± 0.000	0.914 ± 0.002	0.899 ± 0.001
50/50	GB	0.906 ± 0.000	0.928 ± 0.000	0.906 ± 0.001
50/50	KNN	0.897 ± 0.000	1.000 ± 0.000	0.897 ± 0.001
50/50	LR	0.902 ± 0.000	0.902 ± 0.001	0.901 ± 0.000
50/50	RF	0.903 ± 0.000	1.000 ± 0.000	0.903 ± 0.001
50/50	SVM	0.901 ± 0.000	0.946 ± 0.000	0.902 ± 0.001
80/20	GB	0.907 ± 0.000	0.919 ± 0.000	0.907 ± 0.001
80/20	KNN	0.897 ± 0.000	0.908 ± 0.000	0.897 ± 0.001
80/20	LR	0.901 ± 0.000	0.902 ± 0.000	0.900 ± 0.000
80/20	RF	0.903 ± 0.000	0.977 ± 0.001	0.904 ± 0.002
80/20	SVM	0.903 ± 0.000	0.941 ± 0.001	0.903 ± 0.001

Across datasets and partitions. Taken together, the per-partition tables show a consistent pattern: on the more challenging, higher-dimensional, and imbalanced datasets (Spambase, Adult, Bank), **Gradient Boosting and Random Forests are always the top two classifiers**, with gaps of several percentage points over KNN and 1–3 points over the linear and kernel baselines. On the easier Breast Cancer dataset, all methods are near ceiling and the linear/kernel models slightly edge out the ensembles, but the differences are small. This per-dataset view is fully consistent with the aggregate plots and normalized scores, which also rank tree-based ensembles as the strongest overall family of algorithms in this study.

4.2. Per-Classifier, Per-Partition Comparisons

To study how each algorithm benefits from additional labeled data, I average test accuracies over the four datasets and plot them as a function of the training fraction in Figure 1. For every classifier, mean test accuracy increases monotonically from the 20/80 to the 80/20 split, confirming the expected learning–curve behavior.

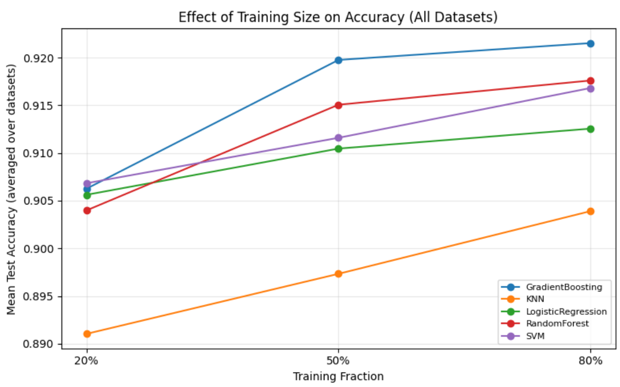

Figure 1. Effect of training size on test accuracy, averaged over the four datasets.

And more specifically, we can observe Per Dataset Behavior in the following figures:

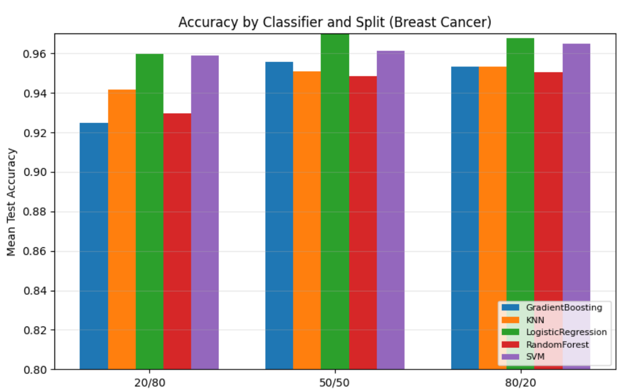

Figure 2. Effect of training size on test accuracy on Breast Cancer Dataset

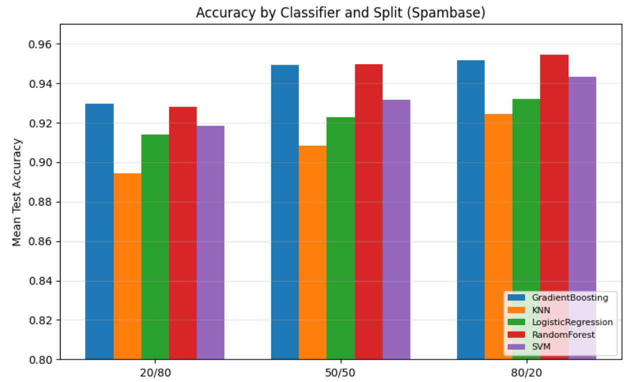

Figure 3. Effect of training size on test accuracy on Spambase Dataset

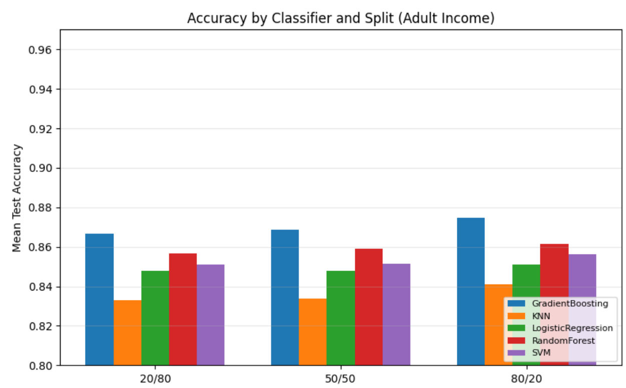

Figure 4. Effect of training size on test accuracy on Adult Income Dataset

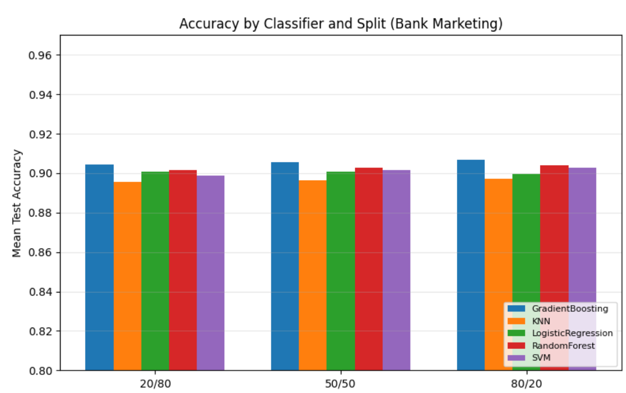

Figure 5. Effect of training size on test accuracy on Bank Marketing

Gradient Boosting. Gradient Boosting consistently achieves the highest or near-highest accuracy on every dataset and partition. On Spambase and Adult Income, its test accuracy increases from roughly 0.93 to 0.95 (Spambase) and 0.87 to 0.88 (Adult) as the train fraction grows from 20% to 80%. Bank Marketing shows a smaller but still positive gain (about 0.904 to 0.907). On Breast Cancer, where all models are already near ceiling performance, accuracy remains very high (≈ 0.95–0.96) with only minor fluctuations. The overall trend in Figure **??** is a clear upward slope, making Gradient Boosting the method that benefits most cleanly from additional data.

Random Forest. Random Forest exhibits very similar behavior. Test accuracy increases with training size on all four datasets, albeit with slightly smaller gains than Gradient Boosting. For example, on Spambase it rises from about 0.93 at 20/80 to 0.95 at 80/20, and on Adult Income from about 0.86 to 0.86–0.87. The aggregate curve in Figure **??** shows a steady improvement from ≈ 0.907 to ≈ 0.913. This pattern is consistent with the CNM finding that tree ensembles are strong, data–efficient classifiers.

Logistic Regression. Logistic Regression forms a middle tier: its absolute accuracies are below the tree ensembles but above k-NN on most datasets. The per-dataset bar plots show modest but consistent accuracy increases with more training data on Adult, Bank, and Spambase. On Breast Cancer it is already very close to 1.0 even at 20/80, so further gains are small and dominated by variance across splits. The overall curve still slopes upward, confirming that additional data helps even this relatively simple linear model.

Support Vector Machine (RBF). SVM with RBF kernel behaves similarly to Logistic Regression but with slightly higher accuracies on some datasets (e.g., Breast Cancer and Spambase). On Adult and Bank, its accuracy increases by roughly 0.004–0.006 from the smallest to largest training fraction. Again, Breast Cancer is near ceiling at all partitions, so the main signal comes from the larger tabular datasets. The aggregate accuracy trend is nearly monotone increasing.

k-Nearest Neighbors. k-NN is consistently the weakest classifier in terms of absolute accuracy, as reflected in both the per-dataset bar plots and the overall curve. However, it still shows clear benefits from more training data. On Spambase, for example, accuracy rises from about 0.89 to 0.92 as the training fraction grows, and on Adult it increases from about 0.83 to 0.84. Bank and Breast Cancer exhibit smaller changes, but the direction is generally upward. This matches the intuition that instance-based methods are data-hungry and remain competitive but rarely best, in line with the CNM conclusions.

Overall, Figures 1-5 together demonstrate that (i) test accuracy almost always improves as the training set becomes larger and the test set smaller for each classifier, and (ii) Gradient Boosting and Random Forests remain the top-performing methods across partitions, while SVM and Logistic Regression form a middle tier and k-NN lags behind but still benefits from additional data.

4.3. Overall performance across datasets

To summarize performance across all four datasets and three train/test partitions, I average each classifier's test metrics over all 180 runs.

Figure 6 shows the ranking by mean test accuracy. Gradient Boosting attains the highest overall accuracy (≈ 0.916), followed very closely by Random Forests, SVMs, and Logistic Regression (all around 0.91); k-NN lags behind with a mean accuracy just under 0.90. The gap between Gradient Boosting and the weaker methods is modest in absolute terms but consistent across datasets and splits.

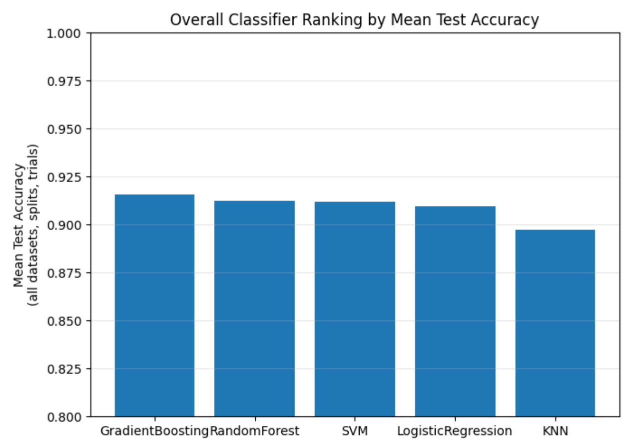

Figure 6. Overall Classifier Ranking by Mean Test Accuracy

The pattern is similar when I consider F_1-score (Figure 7 below). Gradient Boosting again ranks first (mean $F_1 \approx 0.77$), with SVM, Random Forest, and Logistic Regression forming a tight middle tier and k-NN clearly lowest. Because F_1 emphasizes the minority class, especially on Adult and Bank Marketing, these results suggest that the ensemble and margin-based methods handle class imbalance more effectively than k-NN.

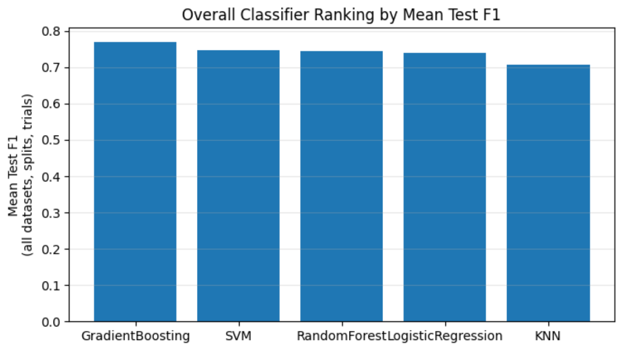

Figure 7. Overall Classifier Ranking by Mean Test F1

For ROC AUC (Figure 8 below), all five methods achieve very strong performance (> 0.93 on average), but Gradient Boosting and Random Forests still lead, with Logistic Regression and SVM only slightly behind and k-NN again last. Taken together, these aggregate metrics reinforce the picture that tree-based ensembles are the most reliable performers across diverse tabular datasets, with linear and kernel methods competitive but slightly weaker, and k-NN the least robust. This qualitative ranking is consistent with the findings of Caruana & Niculescu-Mizil (2006), who also reported that boosted trees and random forests tend to dominate on many benchmark tasks.

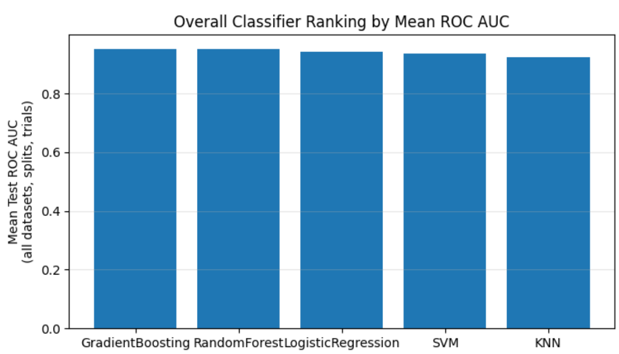

Figure 8. Overall Classifier Ranking by Mean ROC AUC

4.4. CNM-style normalized performance

Raw accuracies and F_1-scores are not directly comparable across datasets with different difficulty. Following Caruana & Niculescu-Mizil (2006), I therefore compute CNM-style normalized scores. For each dataset d I define baseline(d) as the accuracy of a trivial majority-class predictor and best(d) as the highest mean test accuracy achieved by any of the five classifiers on that dataset. The normalized score for classifier c on d is

$$\text{norm\_score}(d, c) = \frac{\text{acc}(d, c) - \text{baseline}(d)}{\text{best}(d) - \text{baseline}(d)}.$$

A score of 0 means the classifier does no better than the majority baseline on that dataset; a score of 1 means it matches the per-dataset winner.

Averaging these normalized scores across the four datasets yields a single CNM-style score per classifier.

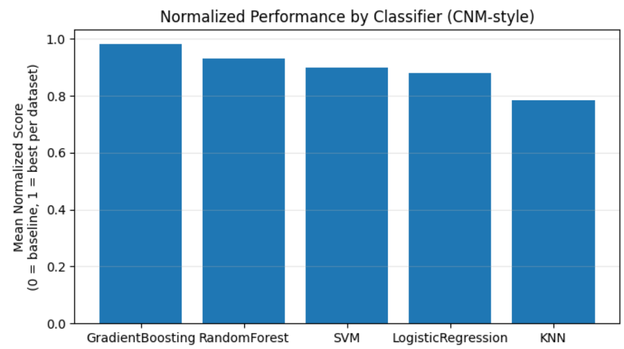

Figure 9. Noramalized Performance by Classifier

Figure 9 plots these values, and Table 6 reports the exact numbers and ranks. Gradient Boosting achieves a mean normalized score of 0.983, meaning that, on average, it recovers almost all of the gap between the majority baseline and the best classifier on each dataset. Random Forests again form a strong second tier (0.931), followed by SVM (0.900) and Logistic Regression (0.880). k-NN is clearly worst with a normalized score of 0.785, indicating that it leaves a substantial portion of the available performance gap unused.

Table 6. CNM-style normalized mean performance across all four datasets. Higher is better.

Classifier	Mean norm. score	Rank
Gradient Boosting	0.983	1
Random Forest	0.931	2
SVM (RBF kernel)	0.900	3
Logistic Regression	0.880	4
k-Nearest Neighbors	0.785	5

The dataset-level averages of the normalized scores also provide a coarse measure of relative difficulty: Spambase and Breast Cancer are moderately challenging (mean normalized scores ≈ 0.61–0.63), Adult Income is somewhat easier (≈ 0.75), and Bank Marketing is easiest overall (≈ 0.88), in the sense that all classifiers come closer to the per-dataset best. Nonetheless, across these very different regimes the ranking of algorithms remains stable, strengthening the conclusion that boosted trees and random forests are consistently strong choices for tabular classification problems.

5. Conclusion and Discussion

In this study I carried out a medium-scale empirical comparison of five standard classifiers on four binary UCI datasets under multiple train/test partitions and cross-validated hyperparameter tuning. Across tables and figures in Section 4, several consistent patterns emerge.

First, tree-based ensembles are the most reliable top performers on these tabular problems. Gradient Boosting achieves the highest overall mean test accuracy, F_1-score, ROC AUC, and the best CNM-style normalized score, with Random Forests a close second. SVM with an RBF kernel and Logistic Regression form a middle tier: they usually outperform k-NN and often come within a few percentage points of the ensembles, especially on the easier Breast Cancer and Spambase datasets. k-NN is consistently ranked last in both raw and normalized performance, but still improves substantially over the trivial majority-class baseline. This mirrors the qualitative findings of Caruana & Niculescu-Mizil (2006), who also reported strong results for boosted trees and random forests and weaker, but not disastrous, performance for k-NN.

Second, training-set size matters in exactly the way standard learning curve intuition would suggest. For every classifier, mean test accuracy (and likewise F_1 and ROC AUC) increases monotonically or near-monotonically as the training fraction grows from 20% to 80%. The gains are modest on Breast Cancer, where many models are already near ceiling at 20% training data, but more pronounced on the larger and more imbalanced Adult and Bank Marketing datasets. The per-dataset bar plots show that the ranking of algorithms is relatively stable across splits, but that all methods benefit from additional labeled examples, with Gradient Boosting and Random Forests typically converting extra data into slightly larger accuracy improvements.

Third, the CNM-style normalization highlights that algorithm rankings are fairly robust across datasets of different difficulty. After rescaling each classifier between the majority-class baseline and the best method on that dataset, Gradient Boosting attains a mean normalized score close to 1.0, indicating that it almost always matches or nearly matches the per-dataset winner. Random Forests follow with a score around 0.93, while SVM and Logistic Regression cluster just below 0.90. k-NN's normalized score of about 0.79 confirms that it is reliably better than baseline but rarely competitive with the strongest methods. This normalized view reduces the impact of absolute dataset difficulty and emphasizes robustness across tasks.

There are several limitations to this work. I restricted attention to four binary tabular datasets and a relatively small family of classical models; modern gradient-boosting implementations (e.g., XGBoost, LightGBM, CatBoost) and neural-network baselines were not included. Hyperparameter grids were intentionally modest to keep computation manageable, so more aggressive tuning or Bayesian optimization might shift some of the fine-grained rankings. Finally, I focused primarily on accuracy, F_1, and ROC AUC; for heavily imbalanced datasets like Bank Marketing, precision–recall curves or cost-sensitive metrics could provide a more nuanced perspective.

Despite these caveats, the overall trends are clear and align well with prior empirical work. On tabular classification problems with mixed feature types, well-tuned tree ensembles remain strong default choices, linear models and kernel SVMs are competitive but somewhat less robust, and k-NN is simple yet rarely state-of-the-art. Extending this analysis to a broader collection of datasets, metrics, and modern algorithms would be a natural next step toward a more comprehensive replication and update of the classic study by Caruana & Niculescu-Mizil (2006).

Bonus Points

Relative to the basic project specification, this work goes beyond the minimum requirements in several ways.

More classifiers and datasets. The specification requires at least three classifiers evaluated on three datasets. Here I study *five* classifiers (Gradient Boosting, Random Forests, SVM with RBF kernel, Logistic Regression, and k-NN) across *four* benchmark UCI datasets (Breast Cancer, Spambase, Adult Income, and Bank Marketing), resulting in $5 \times 4 \times 3 \times 3 = 180$ independent runs. This larger matrix of experiments provides a more stable view of algorithm rankings and better mirrors the spirit of Caruana & Niculescu-Mizil (2006).

Richer evaluation protocol. Beyond the minimal requirement of reporting test accuracy, I record training accuracy, cross-validated validation accuracy, test accuracy, F_1-score, and ROC AUC for every run. I also provide detailed per-dataset, per-partition tables and per-classifier learning curves that explicitly demonstrate how performance changes as the training fraction increases from 20% to 80%. These analyses directly address both requested comparison axes: (a) between algorithms on each dataset and partition, and (b) within each classifier across different train/test partitions.

CNM-style normalization and aggregated analysis. Inspired by Caruana & Niculescu-Mizil (2006), I implement a normalized performance measure that rescales each classifier's accuracy on a dataset between a majority-class baseline and the best observed method on that dataset. I then average these scores across datasets to obtain a single CNM-style score per classifier, together with a summary table and figure. This normalization is not required by the project description, but it allows more principled cross-dataset comparison and shows that Gradient Boosting and Random Forests are consistently strong while k-NN, though weakest, still substantially improves upon the baseline.

Thorough hyperparameter search and documentation. For each (dataset, classifier, partition) combination I perform systematic cross-validated grid search over multiple hyperparameters, using shared preprocessing pipelines and stratified folds. The exact search ranges, CV configuration, and implementation details are documented in the Methods and Implementation sections, and all results are stored in a consolidated results table that drives every figure and summary statistic. The combination of breadth (more models/datasets) and depth (multiple metrics, normalized scores, and extensive visualizations) reflects a level of empirical thoroughness that goes beyond the basic project requirements.

References

Caruana, R. and Niculescu-Mizil, A. An empirical comparison of supervised learning algorithms. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pp. 161–168, Pittsburgh, PA, USA, 2006. ACM. URL https://www.cs.cornell.edu/~caruana/ctp/ct.papers/caruana.icml06.pdf.